GridShift Electricity Demand Forecasting

Phase 3 Model Architecture and Phase 4 Empirical Evaluation

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KEY FINDING

TFT has the lowest reported validation and test errors. LSTM is second and XGBoost third. All models use the same prepared split files; LSTM produces fewer predictions because it requires a 168-row input sequence.

Executive summary

This report covers model architecture, validation-based tuning, held-out testing, error analysis, and the repository contribution record. The target is next-hour electricity demand in MW, reconstructed from the predicted one-hour demand change. The analysis is observational and intended for research, not live grid operation.

Model	Validation MAE / RMSE (MW)	Test MAE / RMSE (MW)
TFT	1,566.80 / 3,316.47	2,985.18 / 6,272.79
LSTM	1,998.05 / 3,762.90	3,567.73 / 6,641.47
XGBoost	3,604.19 / 6,188.10	4,619.62 / 7,930.16

Scores are not exact same-timestamp comparisons: the LSTM evaluation excludes the first 167 feature-ready rows in each split to fill its 168-hour sequence. Prediction counts and this limitation are reported in the evaluation section.

1. Data cleaning and shared split

The model input is **outputs/processed/grid_hourly_merged.csv**, containing 37,392 hourly timestamps from 2021-09-01 00:00 to 2025-12-06 23:00. The current preparation code sorts and deduplicates timestamps, resamples hourly, interpolates internal gaps up to three hours, removes rows with unresolved demand, creates cyclical calendar features, and scales numeric measurements using training-only statistics.

Preparation stage	Rows / counts
Merged source timestamps	37,392
Missing values before interpolation (each of load, solar, wind)	11,714
Missing values after <=3-hour interpolation (each variable)	11,645
Retained rows after unresolved load gaps removed	25,747
Prepared split total	25,747

The earlier 12,984-row summary described a separate workbook. It did not match the source or split files used by these model runs and has been superseded in the cleaning report.

Split	CSV	Rows	Share
Train	train_scaled.csv	18,022	70%
Validation	val_scaled.csv	3,862	15%
Test	test_scaled.csv	3,863	15%

After demand-change, lag, and rolling features, XGBoost/TFT have 17,854 train, 3,694 validation, and 3,695 test rows. The LSTM sequence builder yields 17,687 / 3,527 / 3,528 examples because it needs 168 consecutive feature rows. Its validation predictions begin 167 feature-ready rows after the first eligible validation row; the test has the same boundary effect.

Timestamp continuity: the retained validation split contains 4 gaps and the test split 19 gaps longer than one hour. Because targets shift one retained row, the labels immediately before these gaps refer to the next retained observation, not exactly one clock hour later.

Forecast target and features

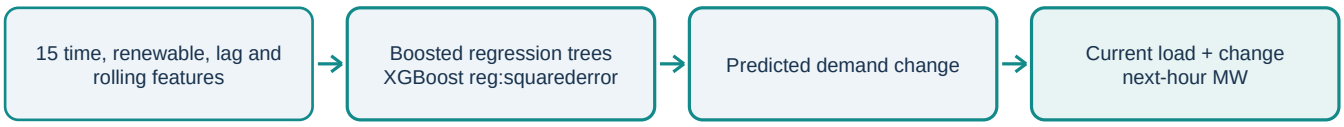
The models predict Δload(t) = load(t+1) - load(t) and reconstruct the MW forecast as load(t) + predicted Δload(t). The 15 input columns are calendar sine/cosine terms, solar, wind, one-, two-, and 24-hour load lags, one-hour solar/wind lags, and 24-/168-hour rolling load means. The split CSVs are loaded via src/data/prepare_data.py.

2. Phase 3 model architecture

Each model solves a continuous one-hour forecasting task. Model design, loss, and framework are documented below.

XGBoost

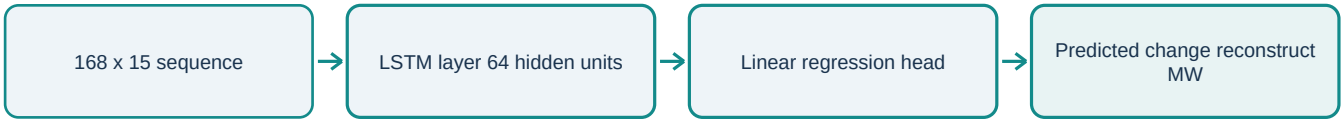
XGBoost's regressor API fits boosted trees to the next-hour demand change with squared-error objective. The selected tree ensemble uses maximum depth 6, learning rate 0.05, 200 trees, subsample 0.8, and column sample 0.8. The current implementation uses 15 forecast-available features.



XGBoost forecast path.

LSTM

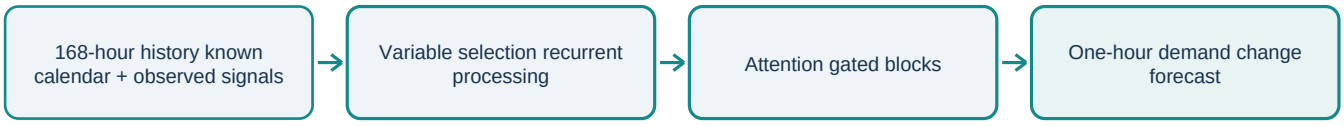
PyTorch implements one LSTM layer with 64 hidden units and a linear regression head. Each input is a 168-hour sequence of 15 features. Training minimizes MSE after training-only standardization of features and target change, with Adam at 0.001 and gradient clipping at 1.0. Validation MAE in MW selects the trial and checkpoint.



Selected LSTM architecture.

Temporal Fusion Transformer

PyTorch Forecasting builds a TFT with a 168-hour encoder, variable selection, recurrent processing, attention, and gated components. Calendar terms are known time features; demand, solar, wind, and target change are historical variables. The selected configuration uses MAE loss, Adam, hidden size 48, four attention heads, hidden continuous size 24, dropout 0.15, learning rate 0.002, and batch size 64.



High-level TFT path.

3. Tuning and learning curves

The validation split selects model settings. Test results are kept for the final evaluation and are not used to choose a configuration.

LSTM settings

Trial	Hidden / layers	Dropout	Learning rate	Batch	Validation MAE (MW)
small	32 / 1	0	0.001	128	2,125.05
balanced (selected)	64 / 1	0	0.001	128	1,998.05
deeper	64 / 2	0.10	0.0005	128	2,090.32

TFT settings

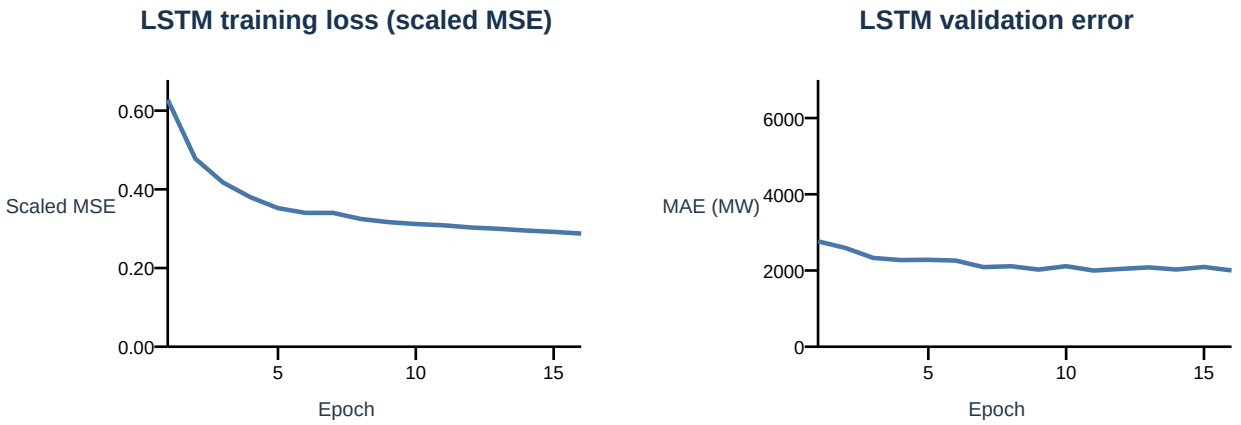
Trial	Hidden	Heads	Hidden cont.	Dropout	LR	Batch	Val MAE
small fast	16	2	8	0.10	0.010	64	1,709.85
balanced	32	4	16	0.10	0.003	64	1,643.16
regularized	32	4	16	0.20	0.001	128	1,666.71
larger (selected)	48	4	24	0.15	0.002	64	1,566.80

XGBoost search

Eight settings cross max_depth {4, 6}, learning_rate {0.05, 0.10}, and n_estimators {200, 400}; subsample and colsample_bytree are held at 0.8. The best validation MAE is 3,604.19 MW (depth 6, 0.05, 200 trees).

XGBoost parameter	Values tested	Selected
max_depth	4, 6	6
learning_rate	0.05, 0.10	0.05
n_estimators	200, 400	200
subsample / colsample_bytree	0.8 fixed	0.8 / 0.8

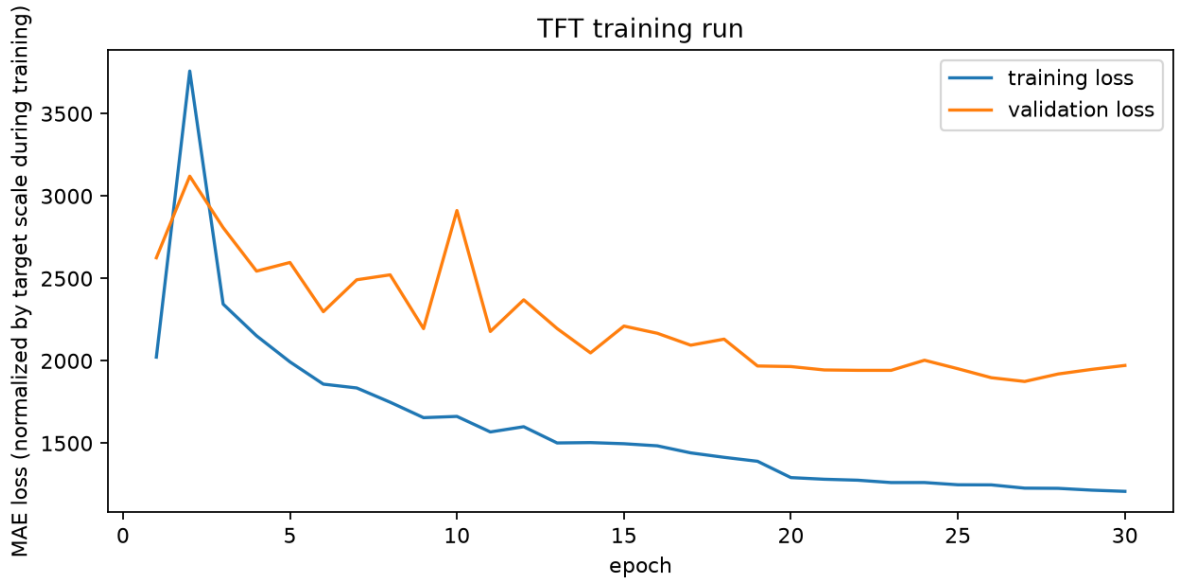
Learning curves



The charts use separate axes and units. The latest LSTM's lowest validation MAE occurred at epoch 11 (1,998.05 MW); early stopping ended training at epoch 16.

TFT training curve

The saved curve tracks training and validation loss across the selected TFT trial. TFT settings were selected by validation MAE; the held-out test set remained reserved for final evaluation.



The curve is included as the TFT learning-curve artifact from the saved run. Exact epoch-by-epoch values are not stored in a separate CSV.

4. Phase 4 results and error analysis

This is continuous-value regression. MAE is the mean absolute error in MW and is directly interpretable. RMSE penalizes large misses more heavily. Precision, recall, F1, and ROC-AUC require discrete class labels or a threshold and do not fit the continuous-demand target. Regression learning curves therefore show training loss and validation MAE rather than classification accuracy.

Model	Split	MAE (MW)	RMSE (MW)	Prediction rows
TFT	Validation	1,566.80	3,316.47	3,694
TFT	Test	2,985.18	6,272.79	3,695
LSTM	Validation	1,998.05	3,762.90	3,527
LSTM	Test	3,567.73	6,641.47	3,528
XGBoost	Validation	3,604.19	6,188.10	3,694
XGBoost	Test	4,619.62	7,930.16	3,695

Ranking

TFT has the lowest reported error on validation and test, LSTM is second, and XGBoost is third. Test MAE is higher than validation MAE for every model: +90.5% for TFT, +78.6% for LSTM, and +28.2% for XGBoost. XGBoost MAPE is 1.93% on validation and 2.42% on test. This indicates the later holdout is harder for the fitted models. Since LSTM uses fewer timestamps, the ranking is informative but not a strict same-row comparison.

Bias and failure checks

For LSTM, mean signed error (prediction minus actual) is -114.50 MW on validation and -208.54 MW on test: slight average underprediction. On test, lowest-demand quartile MAE is 4,365 MW; the model overpredicts that quartile by 2,129 MW on average. In the highest-demand quartile, it underpredicts by 2,058 MW. Hour 23 has the largest LSTM MAE (6,241 MW validation; 12,063 MW test). TFT also has its highest test hourly MAE at hour 23 (12,640 MW).

This check examines demand level and hour of day. No demographic attributes are present, so a demographic fairness test cannot be performed. XGBoost's supplied metrics include validation/test summaries, but its saved prediction artifact is test-only; detailed validation slice analysis is unavailable from that file.

Limitations

The evaluation uses one chronological holdout. The higher later-period errors may reflect temporal distribution shift but do not identify its cause. Results do not establish that renewable generation causes demand changes and should not be used for operational dispatch.

5. Git contributions and reproduction

The contribution mapping uses commit author strings and IDs from the repository history. Git usernames are preserved as recorded and are not treated as verified legal names.

Module	Git author	Relevant commits
Cleaning, splits, shared loader	Abdullah Aftab	1e9fc02 (chronological split/loader); f7f866b (cleaning/data update)
XGBoost implementation and run	Abdullah Aftab	f7f866b
TFT implementation and evaluation	Abdul Muiz	8daf0c4; f6c3cdb
LSTM implementation/results and split alignment	WikidSmellz	7c7db34; 014b6c1
Prior report and README update	WikidSmellz	7c7db34

This report refresh and the related cleaning-report updates are not yet represented in Git history. Commit them under the actual contributor account(s) before submission; this report does not assign those new changes to a prior author.

Reproduction

Install dependencies from requirements.txt and install a PyTorch build that matches the local CPU/GPU. From the repository root, run cleaneddata.py only when you intend to rebuild and overwrite the shared split files, then run src/models/xgboost_model.py, src/models/lstm_model.py, and src/models/tft_model.py. Keep metrics, settings, tuning tables, prediction files, and checkpoints from each run together.

Project files

The editable report source is gridshift_phase1_phase4.tex. The corrected cleaning summary is outputs/processed/audit/cleaning_report.json with a readable companion cleaning_report.md. Model artifacts are under outputs/results/. Current checkpoints are smaller than 100 MB.

References

Mukherjee, D., Kalita, K., & Kumar, S. (2025). *Electricity Demand, Solar and Wind Generation Data (September 2021-June 2025) of India at 1-hour interval*, Version 2. Mendeley Data. DOI: 10.17632/y58jknpgs8.2.
Hochreiter, S. & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735-1780.
Lim, B., Arik, S. O., Loeff, N., & Pfister, T. (2021). Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting. *International Journal of Forecasting*, 37(4), 1748-1764.